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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING DEVOPS LAB PROJECT

CI/CD Pipeline for Deploying Applications to Kubernetes Using Jenkins

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OBJECTIVE :

The main objective of this work is to design, implement, and analyze a **Continuous Integration and Continuous Deployment (CI/CD) pipeline** that automates the complete process of application delivery to a **Kubernetes cluster** using **Jenkins**. The pipeline aims to eliminate manual deployment steps, ensure consistency across different environments, reduce deployment failures, and enable faster and more reliable software releases. By integrating CI/CD practices with container orchestration, the system improves scalability, fault tolerance, and operational efficiency.

ABSTRACT

Continuous Integration and Continuous Deployment (CI/CD) have become fundamental practices in modern software development and DevOps environments. This work presents the design and implementation of a CI/CD pipeline using Jenkins for deploying containerized applications to Kubernetes clusters. Jenkins automates the build, integration, and deployment processes, Docker provides application containerization,

and Kubernetes manages orchestration, scaling, and availability. The pipeline enables frequent and reliable application updates, minimizes human errors, and supports cloud-native deployment models. The proposed system improves deployment speed, application reliability, and infrastructure scalability.

INTRODUCTION

In traditional software development methodologies, application deployment is often performed manually and requires significant coordination between development and operations teams. This approach leads to several challenges, including inconsistent environments, delayed releases, configuration errors, and increased system downtime. As applications grow in size and complexity, these challenges become more severe.

CI/CD pipelines address these issues by automating the integration, testing, and deployment of applications. Jenkins is a widely used open-source automation server that supports the creation of customizable

CI/CD pipelines. Kubernetes is a container orchestration platform designed to manage containerized applications efficiently across distributed clusters. The integration of Jenkins with Kubernetes provides a powerful solution for automated, scalable, and reliable application deployment in modern cloud environments.

HARDWARE AND SOFTWARE REQUIREMENTS :

Hardware Requirements

- Computer system with reliable Internet connectivity
- Minimum **4 GB RAM** (8 GB recommended for Kubernetes clusters)
- Local machine or cloud infrastructure for Kubernetes cluster

Software Requirements

- Jenkins Automation Server
- Docker Engine for containerization
- Kubernetes (Minikube for local setup or cloud platforms such as EKS, AKS, or GKE)
- kubectl Command Line Interface
- Git Version Control System
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EXISTING SYSTEM :

In the existing system, application build and deployment processes are mostly manual or semi-automated. Developers manually build applications, create deployment packages, and deploy them to servers. This approach leads to environment inconsistencies, longer deployment cycles, and frequent human errors. Additionally, scaling applications and handling failures require manual intervention, which increases downtime and operational overhead.

PROPOSED SYSTEM:

The proposed system introduces a **fully automated CI/CD pipeline using Jenkins** to handle application delivery. Whenever a developer commits code to the Git repository, Jenkins automatically triggers the pipeline. The pipeline performs code checkout, build, and testing, followed by Docker image creation. The container image is then deployed to a Kubernetes cluster. Kubernetes ensures application availability by managing replicas, performing rolling updates, and automatically recovering from failures. This system significantly improves reliability, scalability, and deployment speed.

CONCLUSION :

The CI/CD pipeline using Jenkins and Kubernetes provides an effective and automated solution for modern application deployment. By combining CI/CD practices with container orchestration, the system improves software quality, deployment efficiency, and operational scalability.

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